

Caleb Pollreis

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EDUCATION

Bachelor of Science in Computer Engineering (Co-op)

University of Manitoba

- GPA: 3.87
- Internship Availability: Jan. 2027 - Aug. 2027

Expected Graduation Date - April 2028

Winnipeg, Manitoba

SUMMARY

Computer Engineering student interested in **Autonomous Systems**, **Robotics**, and **Embedded Software** roles. Held multiple internships shipping **C++/ROS2** perception on **NVIDIA Jetson**, and currently lead a 10+ member **FSAE** driverless team.

AUTONOMOUS SYSTEMS & ROBOTICS EXPERIENCE

FSAE Electric - Autonomous System Lead (DSO)

Jan. 2026 – Present

Winnipeg, Manitoba

- Leading 10+ members building an autonomous software stack targeting self-driving by 2028, 2 years ahead of schedule
- Architected the **C++/ROS2 Jetson Orin** monorepo for 6 subteams and won **\$30k** in sponsored **LiDARs**/cameras

PTx Trimble - Perception Engineering Intern

May 2026 – Aug. 2026

Winnipeg, Manitoba

- Owned a camera **anomaly detection** machine learning model end-to-end, distilling a **CLIP** teacher into a lightweight **CNN** on a **Jetson Orin**, cutting false positives by 50% and model size by 66% to clear alert spam on retrofit autonomy kits
- Built **ROS** bag triggers and a **C++/Cloudini** pipeline decoding point-cloud and **H.264** data faster than real time

MacDon Industries Ltd. - Software Developer Intern

Jan. 2025 – Aug. 2025

Winnipeg, Manitoba

- Built a computer vision grid running **YOLOv11 segmentation** for crop detection at 89% accuracy, automating manual header adjustments
- Generalized a single-camera **ROS** pipeline to N cameras and added **CAN J1939** cut-height control, freeing operators

FSAE Electric - Firmware System Lead

Aug. 2024 – June 2026

- Led a team of 10+ engineers across five embedded code bases for custom **PCBs**, mentoring safe **embedded C** practices and instituting **code-review** processes, which improved code reliability and accelerated development timelines
- Built **CI/CD** pipelines with **GitHub Actions** and established **CAN2.0** communication across **PCBs**, **BMS**, and an **STM32** motor controller, reducing integration time and increasing overall system reliability

FSAE Electric - Software Team Member

Sept. 2022 – Aug. 2024

- Programmed **STM32** microcontrollers on custom **PCBs** and built a **Python/ADC** tool to capture high-current tab-test data, streamlining data collection and improving analysis efficiency

PROJECTS

Rocket League BOT | 🌐 CPollreis/rocket-league-bot

- Trained an offline bot with **reinforcement learning** via **actor-critic PPO (RLGym, PyTorch)** and custom reward shaping in a simulated **Unreal Engine** physics engine to accelerate training, then 1v1 friends in real time

BMO AI Agent | 🌐 CPollreis/bmo-agent

- Designed an offline multimodal AI agent with a quantized **Qwen 3 LLM** in 2GB on a **Raspberry Pi 5** at 240 tok/s and **Whisper** voice I/O

Personal Portfolio | 🌐 CPollreis/personal-portfolio

- Created a portfolio site (calebpollreis.com) with **Astro**, **React**, **Tailwind CSS**, auto-deployed via **GitHub Actions**

TECHNICAL SKILLS

Programming Languages: C++, C, Python, Verilog, Java, R

CS Fundamentals: Object-Oriented Programming, Data Structures, Algorithms

Machine Learning: PyTorch, NVIDIA TensorRT, CUDA Toolkit, ONNX, Model Deployment, Deep Learning, CNN, YOLO, CLIP, Reinforcement Learning, Machine Learning Evaluation, scikit-learn, Weights & Biases, AWS (EC2, S3, SageMaker)

AI Tools: Claude, Claude Code, Codex, Cursor, Large Language Models (LLMs), Generative AI

Robotics & Computer Vision: ROS2, OpenCV, PPO, NVIDIA Jetson Orin, Eigen, Cloudini

Build & Dev Tools: Docker, CMake, GTest, LLVM, clang-tidy, cpplint, GitHub Actions, Jenkins

Embedded & Protocols: FreeRTOS, STM32Cube, Raspberry Pi, CAN (J1939, CAN2.0), SPI, UART/USART, I2C, TCP/IP

Operating Systems: Linux (Ubuntu, Arch), MacOS, Windows

Version Control & Collaboration: Git, GitHub, Bitbucket, Jira

Front End: JavaScript, TypeScript, React, Astro, Tailwind CSS

AWARDS AND ACHIEVEMENTS

Scholarships & Bursaries: Ron and Dorothy Britton Design Engineering Bursary (2x), W.J. Christie Memorial Bursary in Engineering, Guertin Centennial Scholarship, University of Manitoba Student Union Bursary